

Course: CNT 4419 Secure Coding

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This report describes the implementation of an AES-256 decryption program in Python. The program processes three binary files: ciphertext.bin (which contains the encrypted text), key.bin (which holds the 256-bit encryption key), and iv.bin (which has the initialization vectors used in CBC mode). Using these inputs, the program successfully decrypts the encrypted text and writes the plaintext to paragraph.txt.

The decryption process first reads the three binary files in “rb” mode to ensure the encrypted data is handled correctly. Using the pycryptodome library, an AES cipher is created in CBC mode with the provided key and IV. The ciphertext is then decrypted and the program removes the PKCS7 padding to restore the original plaintext. Finally, the decrypted text is saved in paragraph.txt using UTF-8 encoding.

During the implementation, we encountered and solved some difficulties. One of the problems was incorrect padding, which led to incorrect decryption results. This was solved by correctly applying the unpad() function to remove the PKCS7 padding. In addition, the mishandling of files initially led to read errors, which were resolved by ensuring that all input files were opened in binary mode.

The final implementation was successful and paragraph.txt contains the correct English text extracted from the encrypted file. The AES-256 decryption process in CBC mode was applied correctly and the accuracy of the output was verified. This project demonstrates the practical application of symmetric encryption and decryption in a controlled environment.